

CHAPTER 3

IMPLEMENTATION AND DESIGN OF SCHOOL LIBRARY MANAGEMENT SYSTEM

This paper presents school library management system by using C# programming language in ASP.NET Core framework to manage school library circulation and book catalogues. This system allows admins and users to view the book information and circulation records. In this system, admins only can manage book details, categories, members, borrow and return transactions, and system settings. Admins only can manage users.

3.1 User Interface Design



Figure 3.1 User Interface for Home Page Window

When the system starts, the user will see the home page as shown in Figure 3.1. Any user can browse available books, search the catalogue, or click the "Login" button to login.

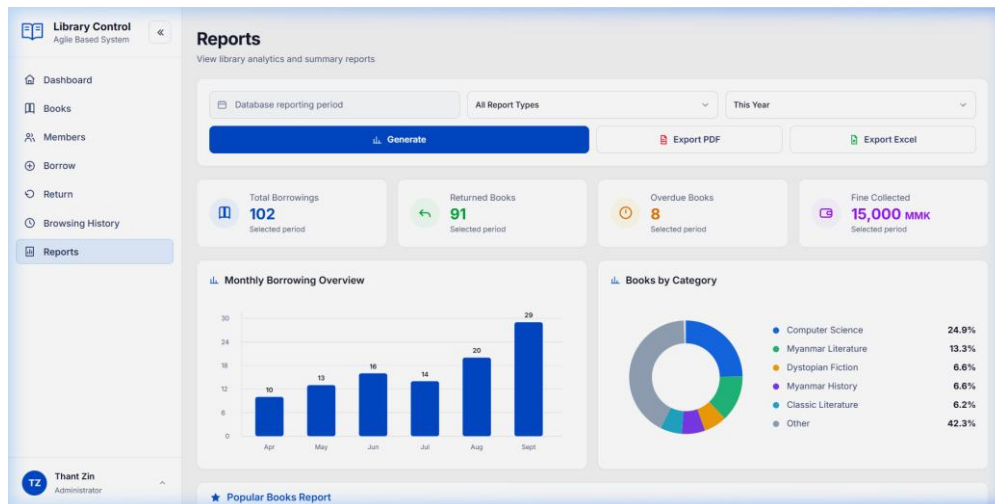


Figure 3.2 Displaying the Browse Books Window

Figure 3.2 shows the browse books catalogue window. In this window, users can explore available book titles, view real-time availability badges, and filter books by category.

Figure 3.3 shows the member self-service borrowing history window. The window provides users with the search tool to lookup their personal borrowing records, due dates, and fine balances.

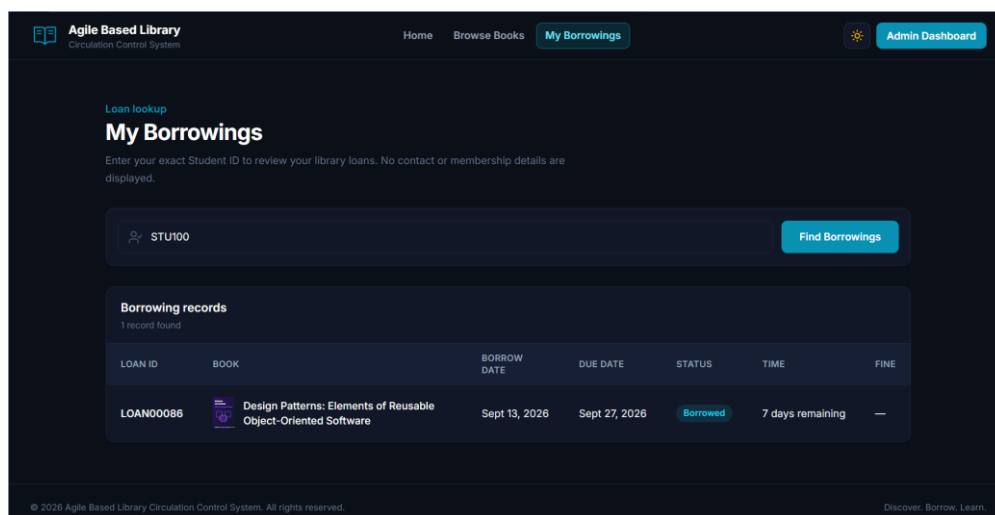


Figure 3.3 Displaying the Member Borrowing Lookup Window

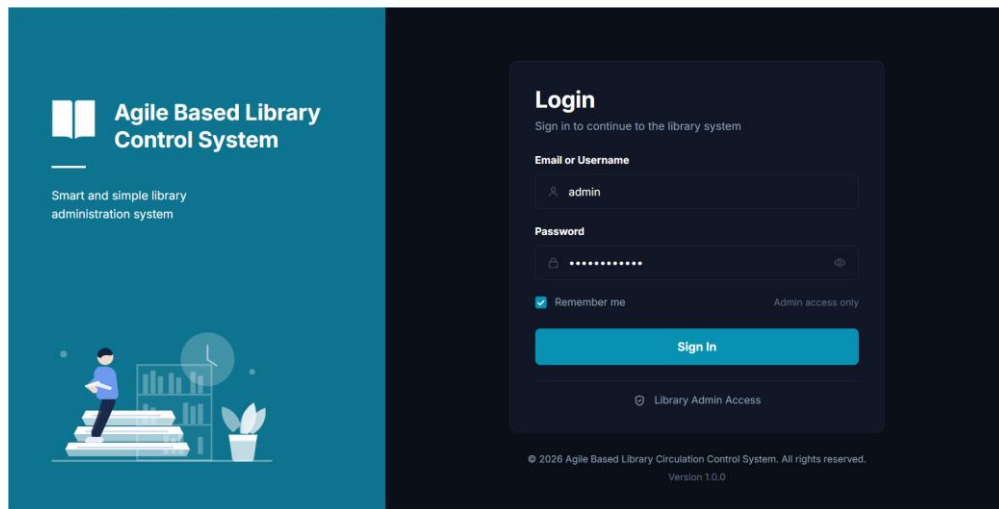


Figure 3.4 Displaying the User/Admin Login Window

Figure 3.4 shows the User/Admin login window. Admin and users can login by filling username/email and password in the form to authenticate into the system.

Figure 3.5 shows the logged in admin home page window. The window provides admin with the navigation bar with Home, Books, Members, Borrow, Return, Reports, Settings and Logout buttons.

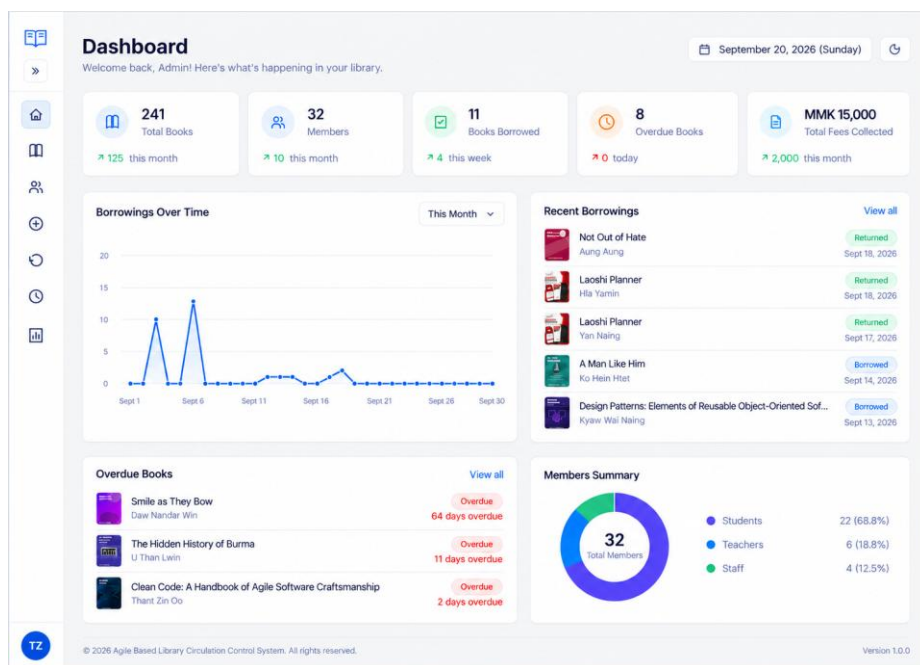
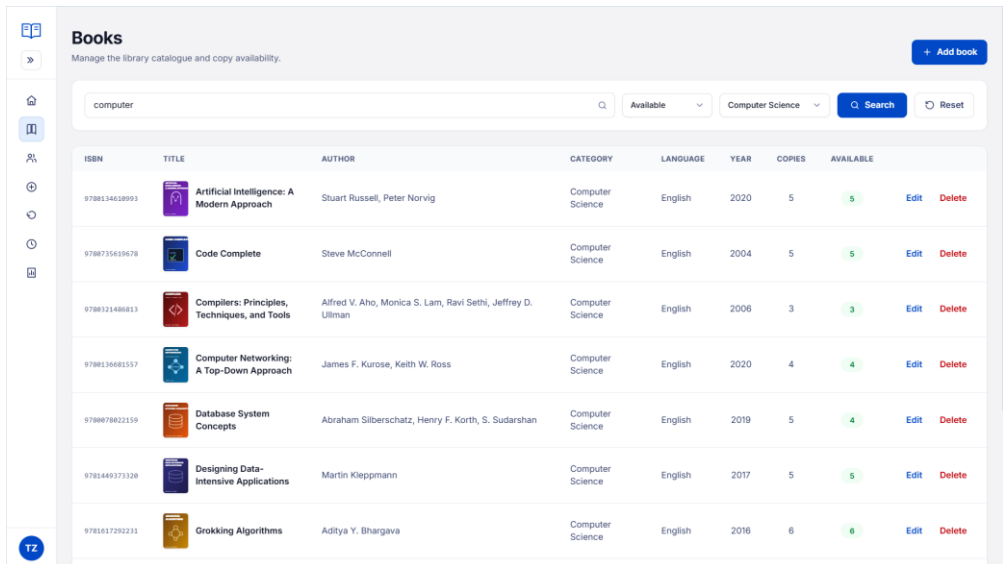


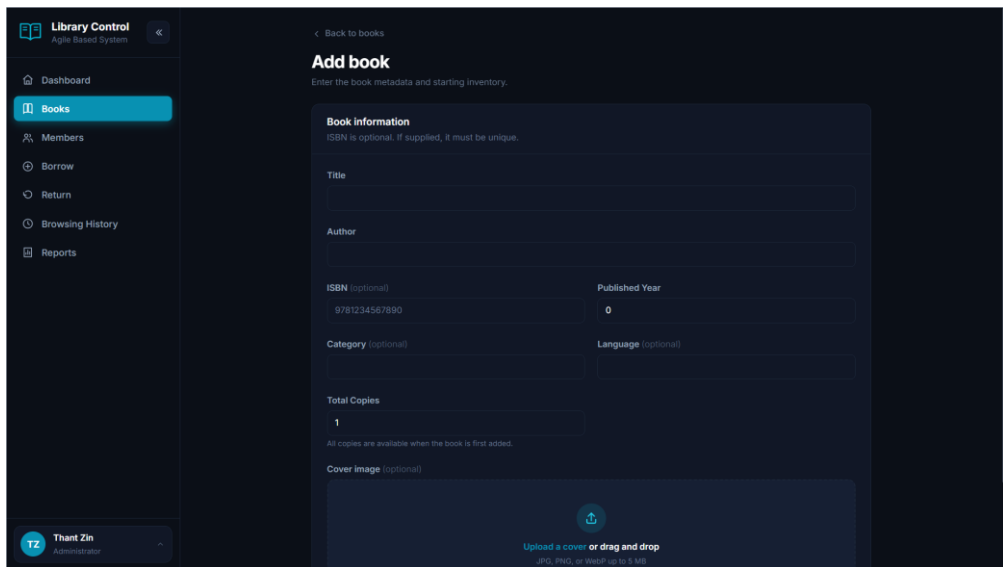
Figure 3.5 Displaying the Admin's Home Page Window

After clicking on the "Books" button, the user will be taken to the "Book Catalogue" window as shown in Figure 3.6. User can see the list of books and search by title, author, category or ISBN from the search bar.



ISBN	TITLE	AUTHOR	CATEGORY	LANGUAGE	YEAR	COPIES	AVAILABLE	
9780134618993	Artificial Intelligence: A Modern Approach	Stuart Russell, Peter Norvig	Computer Science	English	2020	5	5	Edit Delete
9780735619678	Code Complete	Steve McConnell	Computer Science	English	2004	5	5	Edit Delete
9780321486813	Compilers: Principles, Techniques, and Tools	Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffrey D. Ullman	Computer Science	English	2006	3	3	Edit Delete
9780136681557	Computer Networking: A Top-Down Approach	James F. Kurose, Keith W. Ross	Computer Science	English	2020	4	4	Edit Delete
9780078022159	Database System Concepts	Abraham Silberschatz, Henry F. Korth, S. Sudarshan	Computer Science	English	2019	5	4	Edit Delete
9781449373328	Designing Data-Intensive Applications	Martin Kleppmann	Computer Science	English	2017	5	5	Edit Delete
9781017292231	Grokking Algorithms	Aditya Y. Bhargava	Computer Science	English	2016	6	6	Edit Delete

Figure 3.6 Displaying the Book Catalogue Window



Add book

Enter the book metadata and starting inventory.

Book information

ISBN is optional. If supplied, it must be unique.

Title

Author

ISBN (optional)

Published Year

Category (optional)

Language (optional)

Total Copies

1

All copies are available when the book is first added.

Cover image (optional)

Upload a cover or drag and drop

JPG, PNG, or WEBP up to 5 MB

Figure 3.7 Displaying the Add New Book Window

Figure 3.7 shows the add new book modal window. Admin can register new books by filling title, author, ISBN, category, publisher, publish year, total copies, and uploading a cover image.

Figure 3.8 shows the library members administration window. The admin can view all registered members across Student, Teacher, and Staff tiers, search member records, and check active loan counts.

#	ID	NAME	EMAIL	PHONE	TYPE	JOIN DATE	STATUS	ACTION
1	STA1001	Ko Hein Htet	hein.htet@school.edu	09 780 110 205	Staff	Sept 05, 2026	Active	[Edit] [Delete]
2	TEA1001	Daw Nandar Win	nandar.win@school.edu	09 780 110 204	Teacher	Sept 05, 2026	Active	[Edit] [Delete]
3	STU0002	Thant Zin Oo	—	09 999 888 999	Student	Sept 04, 2026	Active	[Edit] [Delete]
4	STU1002	Min Khant Kyaw	min.khant@school.edu	09 780 110 202	Student	Sept 04, 2026	Active	[Edit] [Delete]
5	STU1003	Su Myat Noe	su.myat@school.edu	09 780 110 203	Student	Sept 04, 2026	Active	[Edit] [Delete]
6	STU1001	Thazin Hlaing	thazin.hlaing@school.edu	09 780 110 201	Student	Sept 03, 2026	Active	[Edit] [Delete]
7	STU0017	May Myat Noe	may.myat@school.edu	09 970 654 987	Student	Aug 09, 2026	Active	[Edit] [Delete]
8	STU0016	Noah Taylor	noah.taylor@school.edu	09 782 321 654	Student	Aug 01, 2026	Active	[Edit] [Delete]
9	STA0003	Daw Khin Mar	khin.mar@school.edu	09 422 789 456	Staff	Jul 24, 2026	Active	[Edit] [Delete]
10	STU0015	Min Thura	min.thura@school.edu	09 961 456 123	Student	Jul 17, 2026	Active	[Edit] [Delete]

Figure 3.8 Displaying the Members Administration Directory Window

Add New Member
Enter member details and membership classification.

Member Information
Student ID is required and must be unique.

Student ID * Full Name *

Membership Type * Department (optional)

Phone (optional) Email (optional)

☒ Active member (allowed to borrow books)

Figure 3.9 Displaying the Add New Member Registration Window

Figure 3.9 shows the new member registration modal window. The admin can register new patrons by providing their full name, member code, membership tier, department, contact number, and email address.

Figure 3.10 shows the book borrowing circulation management window. Librarians can view active checkouts, search by member code or transaction ID, monitor return due dates, and track overdue items.

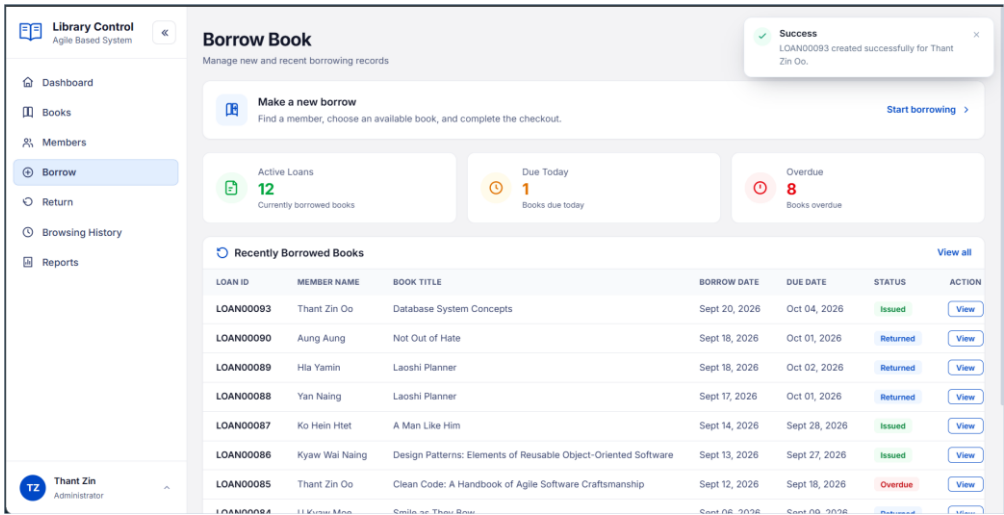


Figure 3.10 Displaying the Book Borrowing Circulation Window

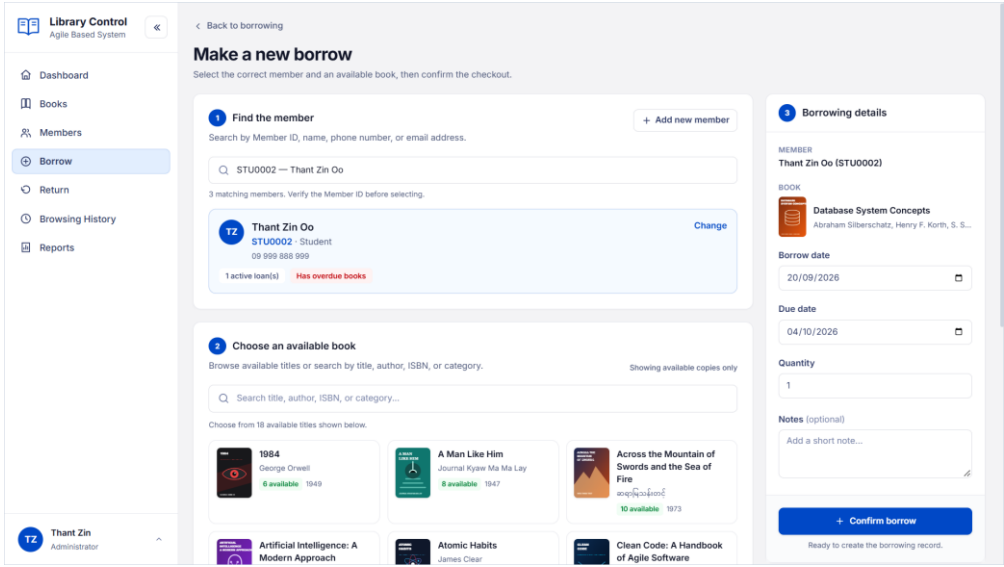


Figure 3.11 Displaying the New Book Checkout and Issue Window

Figure 3.11 shows the checkout window for issuing books to members. The librarian selects an active member, selects one or more books, specifies the due date, and confirms the checkout transaction.

Figure 3.12 shows the book returns directory window. The admin can review all completed book returns, return timestamps, condition gradings, and recorded fine payments.

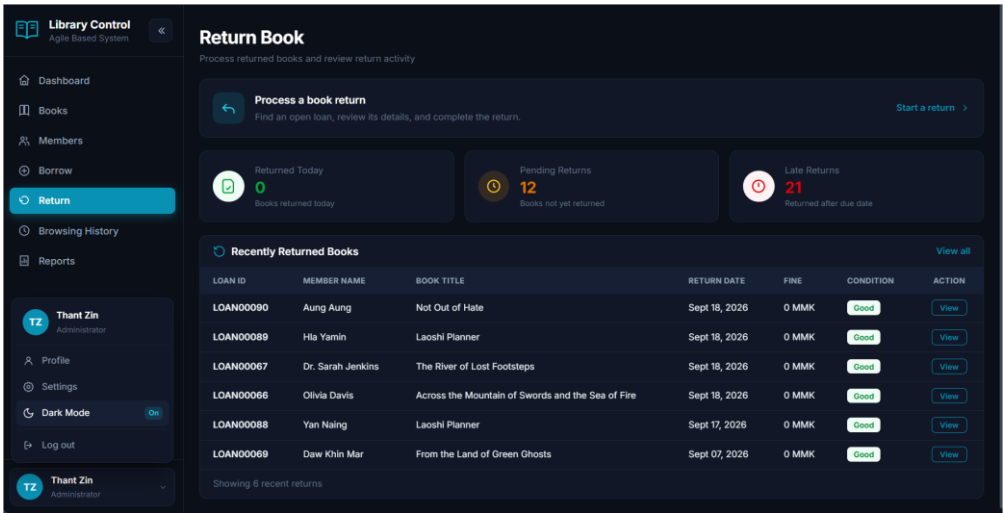


Figure 3.12 Displaying the Book Return Directory Window

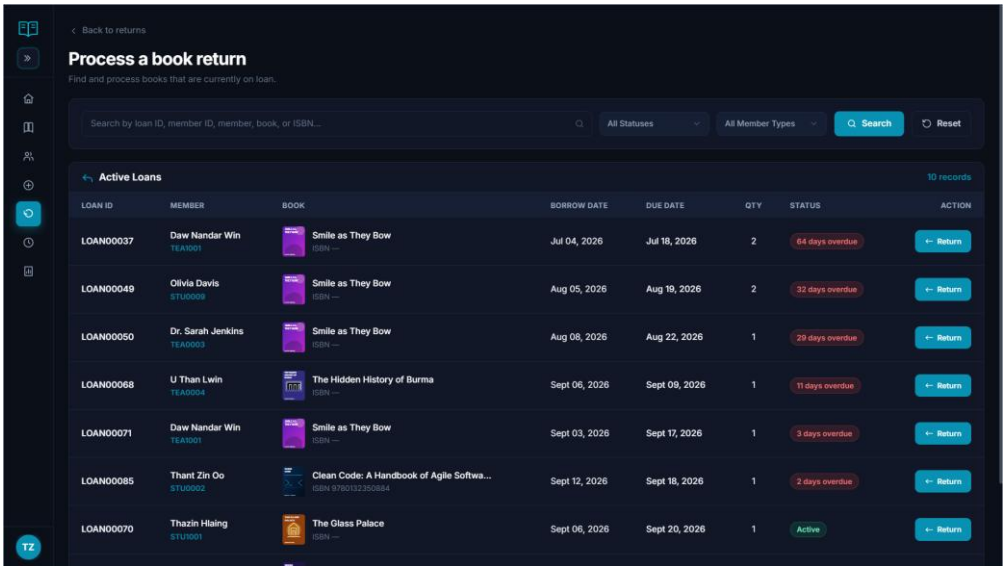


Figure 3.13 Displaying the Process Return and Fine Assessment Window

Figure 3.13 shows the process return modal dialog. The librarian inspects the returned book, records condition notes (Good, Damaged, or Lost), checks overdue days, and collects any assessed overdue fines.

Figure 3.14 shows the circulation borrowing history window. This window provides a comprehensive audit trail of all historical circulation transactions with date range filters.

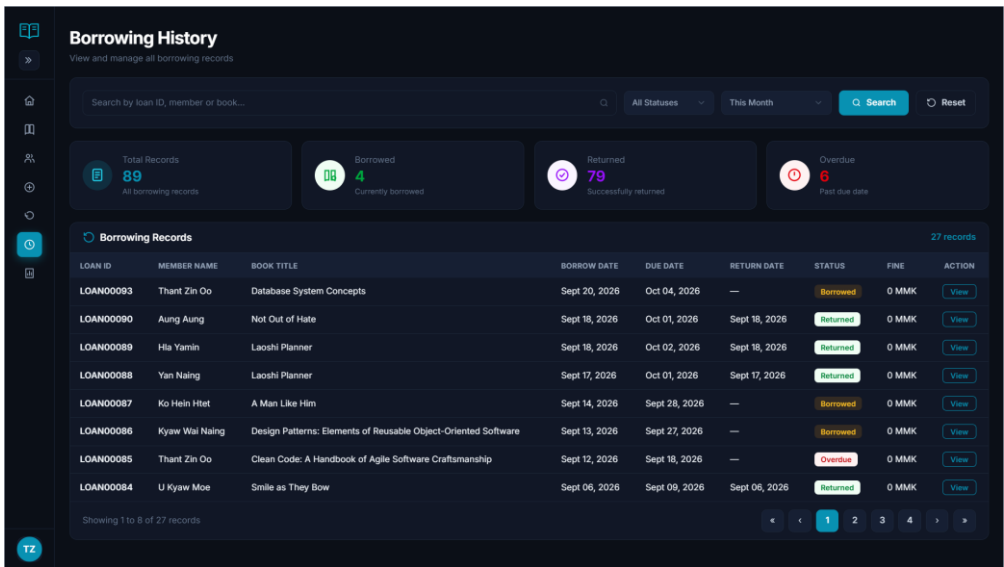


Figure 3.14 Displaying the Circulation History Audit Window



Figure 3.15 Displaying the Analytical Reports and Periodic Statistics Window

Figure 3.15 shows the analytical reports window. Admins can generate circulation performance statistics, overdue summaries, and top-borrowed book rankings filtered by month, year, or custom date ranges.

Figure 3.16 shows the administrator profile window. The logged-in admin can view their profile details, assigned system role, registered email address, and last login activity.

The screenshot shows the 'Library Control' application interface. On the left is a sidebar with navigation links: Dashboard, Books, Members, Borrow, Return, Browsing History, and Reports. The main content area is titled 'Profile' with the subtitle 'Manage the library administrator account details.' It features a user card for 'Thant Zin' (Administrator) with a 'Change password' button. To the right is a 'Personal Information' form with fields for Full name (Thant Zin), Role (Administrator), Email (admin@school.edu), Phone, Department (Library Services), and Bio. The form has 'Cancel' and 'Save profile' buttons at the bottom right. A user status bar at the bottom left shows 'Thant Zin Administrator'.

Figure 3.16 Displaying the Administrator Account Profile Window

The screenshot shows the 'Library Control' application interface with the 'Change Password' window open. The sidebar is the same as in Figure 3.16. The main content area is titled 'Change Password' with the subtitle 'Protect the administrator account with a strong, unique password.' It contains an 'Account security' section with instructions: 'Use at least 10 characters with uppercase, lowercase, a number, and a symbol.' Below this are three input fields: 'Current password', 'New password', and 'Confirm new password'. At the bottom right of the form are 'Cancel' and 'Change password' buttons. The user status bar at the bottom left shows 'Thant Zin Administrator'.

Figure 3.17 Displaying the Password Rotation and Security Window

Figure 3.17 shows the password rotation window. Admins and staff can update their login password with mandatory complexity requirements to ensure system account security.

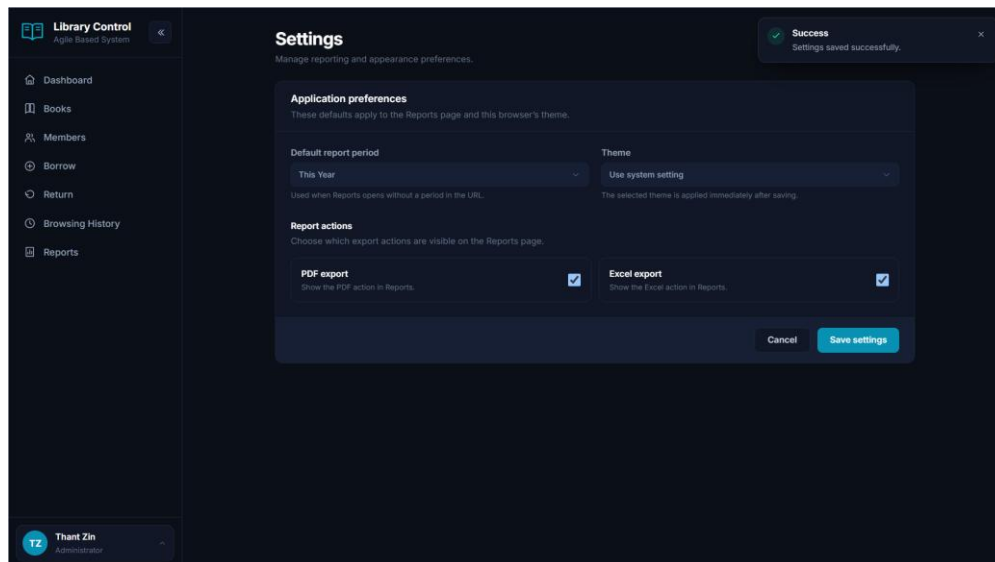


Figure 3.18 Displaying the Global System Settings Window

Figure 3.18 shows the global library settings window. The admin can configure system-wide rules including loan period durations, maximum borrow allowances, daily fine rates, and notification preferences.

3.2 Summary

In this chapter, the user interface design and operational workflows of the School Library Management System were presented. The interface provides accessible and responsive interactions for both public patrons and administrative personnel, ensuring efficient circulation management, inventory tracking, and institutional reporting.